

CheckPlot Statistics: Beta Perturbations

Ranging ϵ using `seq(-0.1, 0, 0.025)` to observe how small perturbations to *uniform* p-values impact the statistics. This is using the *modified* Location statistic that has not yet been formally discussed.

```
empDist.beta <- function(numSims, numReps, epsilon){

  stat.space <- c()
  stat.location <- c()

  for (x in 1:numReps){
    beta.dat <- data.frame(lambda = 0, p=rbeta(numSims, 1+epsilon, 1))
    stats <- checkplotStats(dat = beta.dat, binwidth = 0.01, 0)
    stat.space <- c(stat.space, stats$var2[[1]])
    stat.location <- c(stat.location, stats$var3[[1]])
  }

  return(list("stat.space" = stat.space, "stat.location" = stat.location))
}

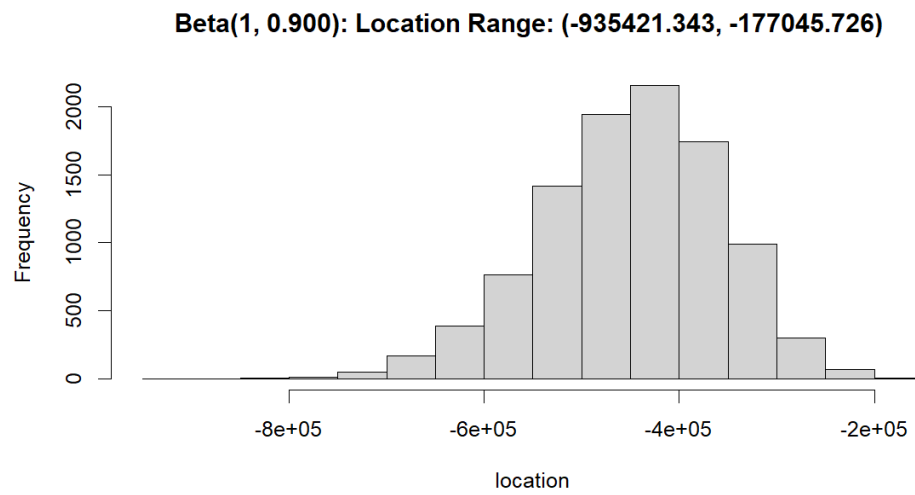
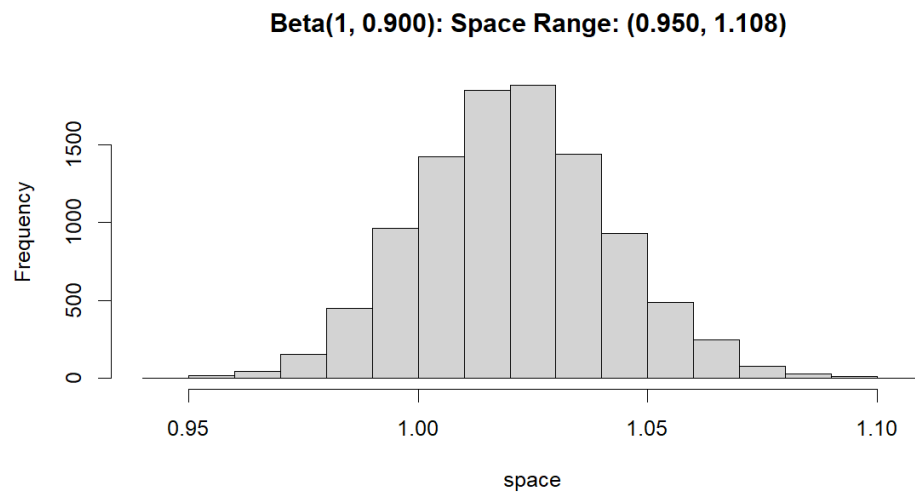
for (e in seq(-0.1,0,0.025)){
  betaStats <- empDist.beta(1e4, 1e4, e)

  space <- betaStats$stat.space
  location <- betaStats$stat.location

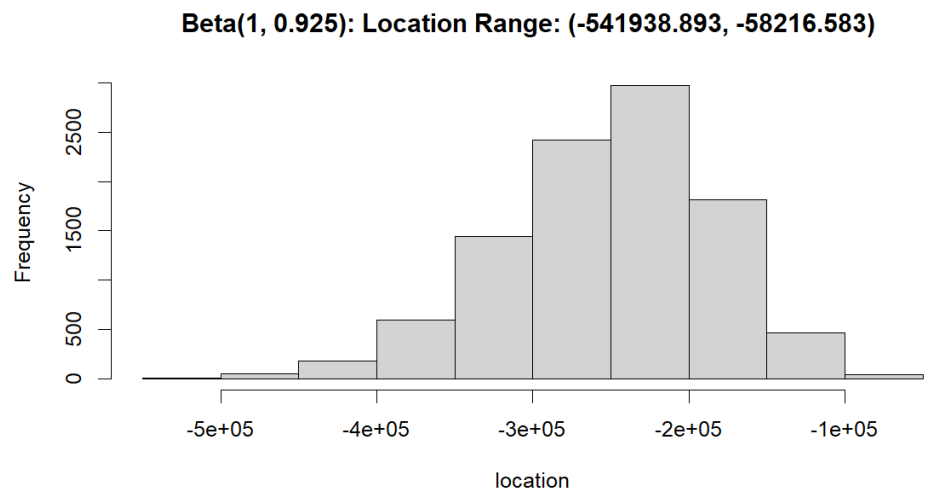
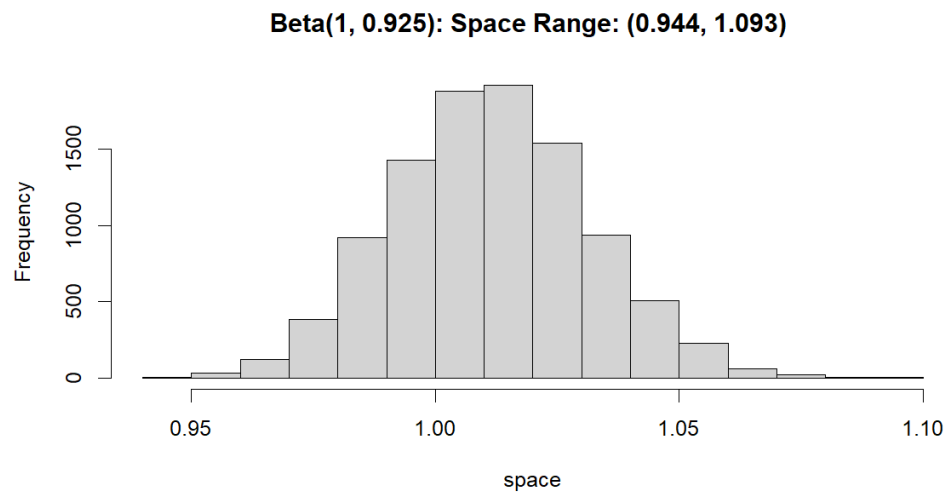
  min <- min(space)
  max <- max(space)
  title <- paste0(sprintf("Beta(%.0f, %.3f): Space Range: (%.3f, %.3f) ",
                          1, 1+e, min, max))
  hist(space, main = title)

  min <- min(location)
  max <- max(location)
  title <- paste0(sprintf("Beta(%.0f, %.3f): Location Range: (%.3f, %.3f) ",
                          1, 1+e, min, max))
  hist(location, main = title)
}
```

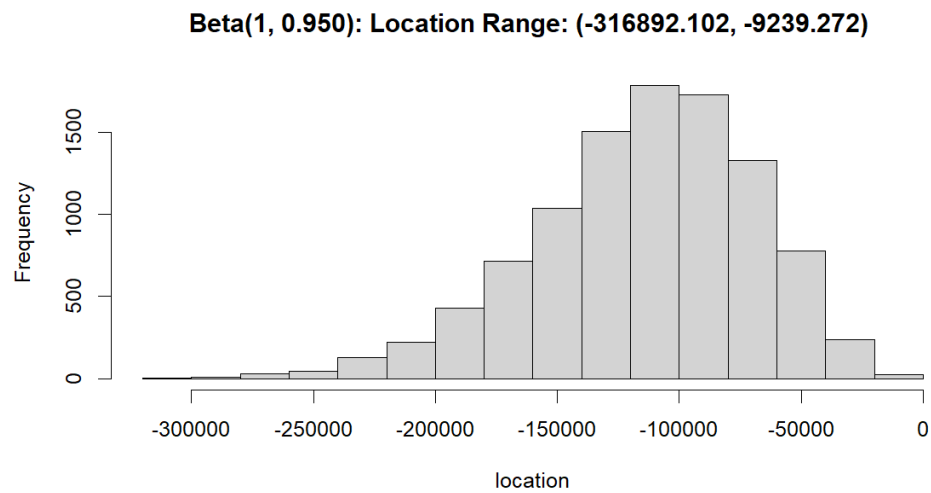
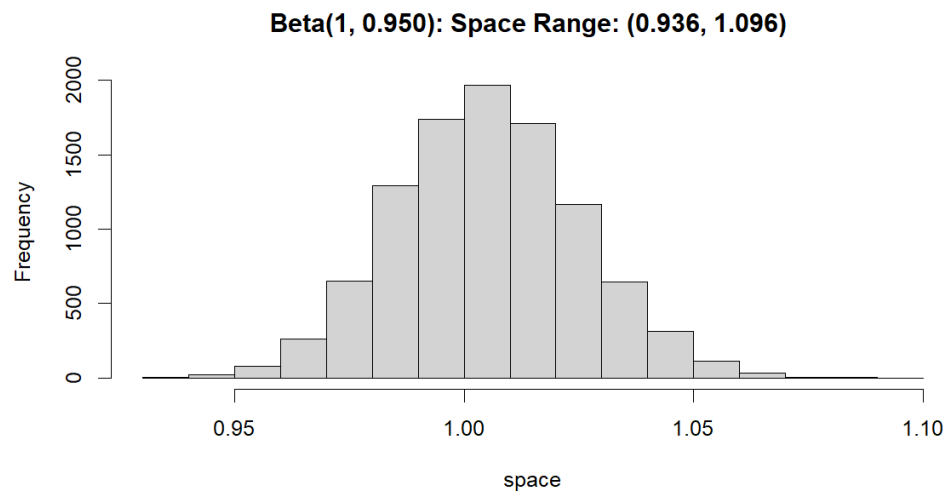
Beta(1, 0.90)



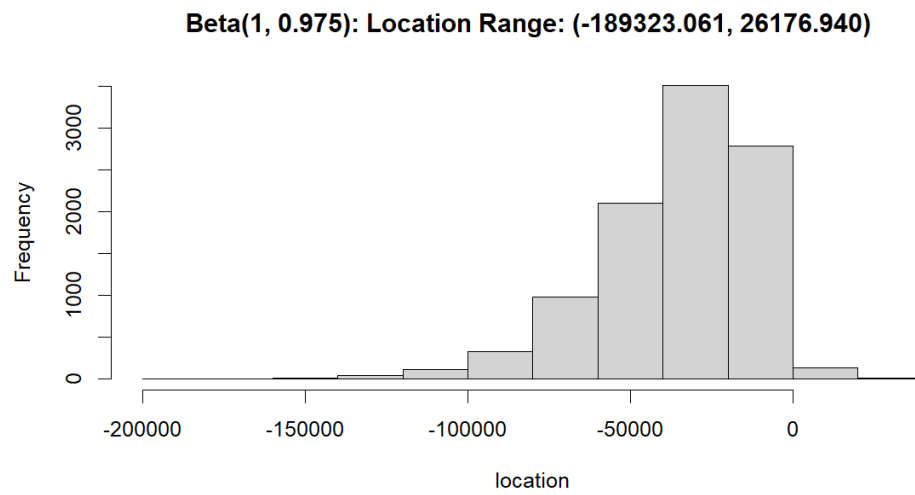
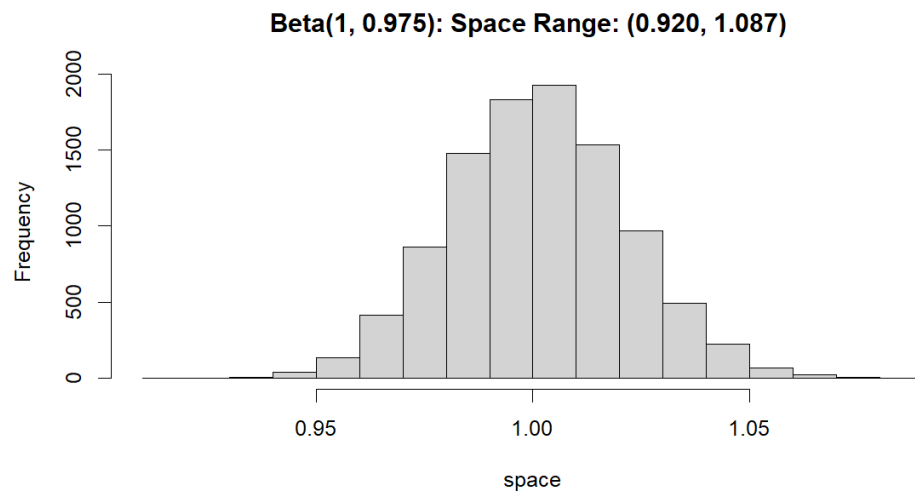
$\text{Beta}(1, 0.925)$



Beta(1, 0.95)



$\text{Beta}(1, 0.975)$



Beta(1,1)

